

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_openml
from sklearn.model_selection import train_test_split
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LinearRegression

boston = fetch_openml(data_id=531, parser='auto')
data = pd.DataFrame(boston.data)
data.columns = boston.feature_names
data['PRICE'] = boston.target

imputer = SimpleImputer(missing_values=np.nan, strategy='mean')
x = pd.DataFrame(imputer.fit_transform(data.drop(['PRICE'],
axis=1)), columns=data.drop(['PRICE'], axis=1).columns)
y = data['PRICE']
data = data.dropna()
x = x.loc[data.index]

xtrain, xtest, ytrain, ytest = train_test_split(x, y,
test_size=0.2, random_state=0)

lm = LinearRegression()
model=lm.fit(xtrain, ytrain)

ytrain_pred = lm.predict(xtrain)
ytest_pred = lm.predict(xtest)

df=pd.DataFrame(ytrain_pred,ytrain)
df=pd.DataFrame(ytest_pred,ytest)

from sklearn.metrics import mean_squared_error, r2_score
mse = mean_squared_error(ytest, ytest_pred)
print(mse)
mse = mean_squared_error(ytrain_pred,ytrain)
print(mse)

33.44897999767632
19.32647020358573

mse = mean_squared_error(ytest, ytest_pred)
print(mse)

33.44897999767632

plt.scatter(ytrain ,ytrain_pred,c='blue',marker='o',label='Training
data')
plt.scatter(ytest,ytest_pred ,c='lightgreen',marker='s',label='Test
data')
plt.xlabel('True values')
plt.ylabel('Predicted')

```

```
plt.title("True value vs Predicted value")
plt.legend(loc= 'upper left')
#plt.hlines(y=0,xmin=0,xmax=50)
plt.plot()
plt.show()
2
```

